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Unit 1 - Exploring the Core

PDF Documents

PDF Document 1: 1738069966854-Chapter 6 - Unit 1 - Exploring the Core.pdf

This PDF document is included in the unit materials.



UNIT 1 EXPLORING THE CORE

Topic Investigation



An earthquake is a natural disaster caused by the sudden release of energy in the Earth's crust. It often leads to the shaking of the ground, resulting in various degrees of damage to buildings, infrastructure, and communities. Understanding the causes of earthquakes and the methods used to detect them is crucial for mitigating their impact and ensuring public safety.

CAUSES OF EARTHQUAKES

The primary cause of earthquakes is the tectonic forces resulting from the movement of these plates. Earthquakes typically occur in three types of settings:

CONVERGENT BOUNDARIES

Where plates move toward each other, often causing one plate to sink into the mantle beneath another.

DIVERGENT BOUNDARIES

Where plates move apart, magma rises from the mantle, creating new crust.

TRANSFORM BOUNDARIES

Where plates slide past each other horizontally, leading to friction and intermittent release of stress.

Minor earthquakes can also occur due to volcanic activity, landslides, or human activities like mining or reservoir-induced seismicity.



HOW TO PREPARE FOR AN EARTHQUAKE

Preparing for an earthquake is crucial, especially in areas prone to seismic activity. Some key preparation steps include:

1 Creating an Emergency Plan:

Ensure that all family members know what to do during and after an earthquake.
Practice regular drills.

2 Building an Emergency Kit:

This should include supplies like water, food, first aid materials, flashlights, and a radio.

3 Securing Heavy Furniture and Objects:

Prevent injuries and damage by anchoring furniture and securing items that might fall.

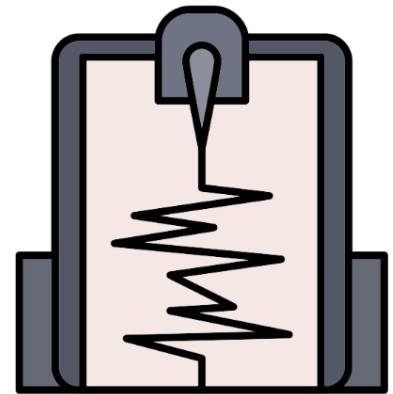


DETECTION AND MEASUREMENT OF EARTHQUAKES

The detection and measurement of earthquakes involve several methods and technologies:

SEISMOGRAPHS

Instruments that measure and record the details of an earthquake, such as force and duration.



IMU SENSORS

Used in modern applications, these sensors can detect changes in rotational attributes and vibrations, making them useful for developing personalized earthquake detection systems like the MODI creation.

EARLY WARNING SYSTEMS

Utilize a network of sensors to provide advance warning of an earthquake, potentially saving lives by alerting people before the shaking starts.



Inquiry & Reflection

Discussion question 1

How would you design your home or classroom to be safer in the event of an earthquake? Consider the location of furniture, the type of building materials used, and any safety equipment that might be helpful.

Discussion question 2

Imagine your school is located in a high-risk earthquake zone. What emergency preparations and drills would you suggest to help students feel more prepared and safe? Discuss practical steps and strategies that could be implemented to enhance safety during an earthquake.